

Choice

JEB104 Microeconomics I

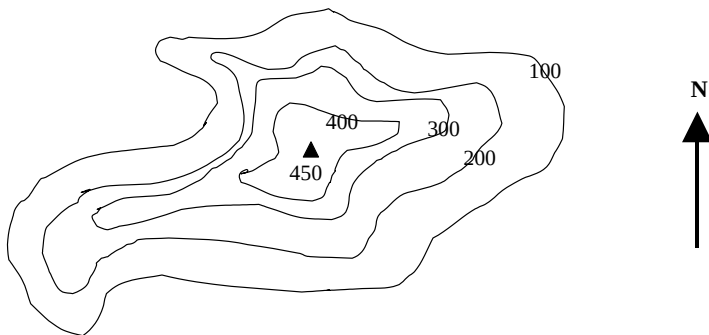
Lenka Št'astná

Prague, IES FSV UK, March 2, 2020

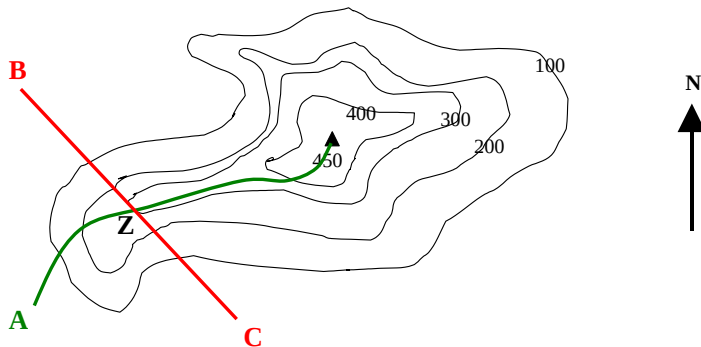
Example of Constrained Maximization

Imagine a hill with height shown by contours

You want to climb as high as possible

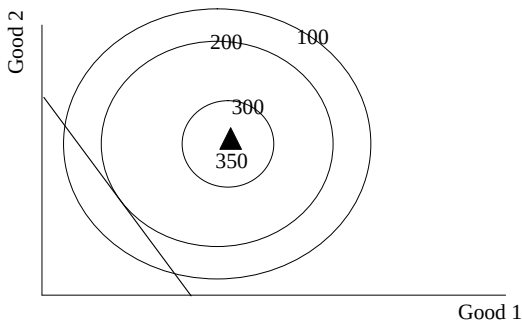


Example of Constrained Maximization (Continued)



- A path runs up the hill from point A to the summit.
- But a fence BC runs across the hill barring access to the public.
- This means that you cannot climb higher up the hill than point Z on the 300 contour line.

Example of Constrained Maximization (Continued)



- Now instead of a physical hill that you climb, imagine a mountain of pleasure, with pleasure measured in terms of utility.
- The top is unreachable, because you have limited income.
- This is the basic idea of how we model consumer behavior.

Optimal Choice

Consumer chooses the best bundle $\mathbf{x}^* = (x_1^*, x_2^*, \dots, x_n^*)$

Constrained to bundles $\mathbf{x} = (x_1, x_2, \dots, x_n)$ in the budget set:

$$B = \{\mathbf{x} \in \mathbb{R}_{+,0}^n : \sum_{i=1}^n p_i x_i \leq m\}$$

Objective function: $u(x_1, x_2, \dots, x_n)$

Optimization problem:

$$\begin{aligned} & \max_{x_1, \dots, x_n} u(x_1, \dots, x_n) \\ \text{s.t.} \quad & \text{(i)} \quad \sum_{i=1}^n p_i x_i \leq m \\ & \text{(ii)} \quad x_i \geq 0 \end{aligned}$$

Optimal Choice

In optimum:

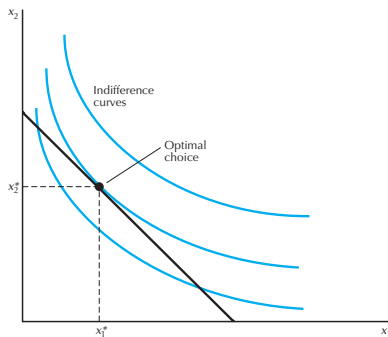
- (i) Consumer spends his entire income:

$$\sum_{i=1}^n p_i x_i^* = m$$

$\Rightarrow \mathbf{x}^*$ on the upper boundary of B (from monotonicity)

- (ii) Out of the bundles on the upper boundary of B , consumer chooses the one lying on the highest indifference curve.

Optimality Condition



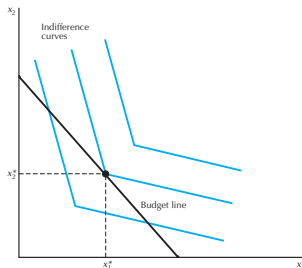
In interior optimum:

$$MRS_{21}(x^*) = -\frac{p_1}{p_2}$$

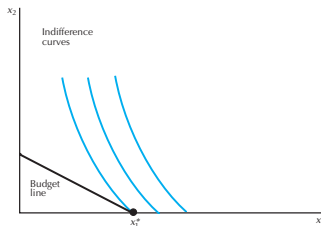
$$-\frac{MU_1(x^*)}{MU_2(x^*)} = -\frac{p_1}{p_2} \Rightarrow \frac{MU_1}{p_1} = \frac{MU_2}{p_2}$$

Optimality Condition May Not Hold in Optimum

1. Kinky tastes: optimal point = kink point (no tangent)

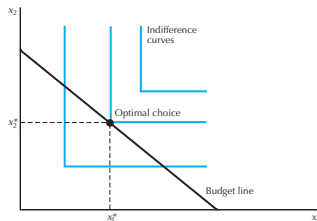


2. Boundary optimum (corner solution)

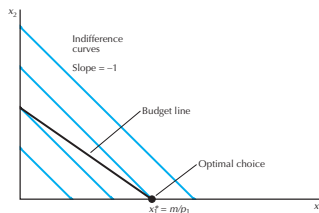


Optimality Condition May Not Hold in Optimum: Examples

Perfect complements

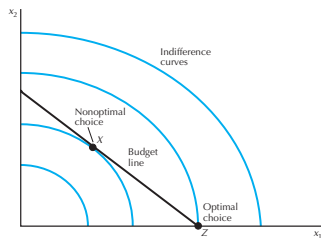


Perfect substitutes



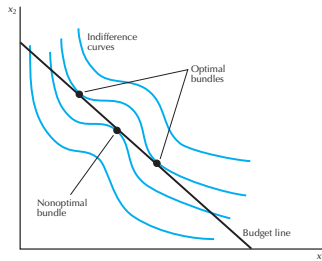
Optimality Condition May Not Hold in Optimum: Examples

Extremely non-convex (“concave”) preferences

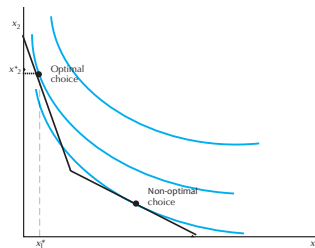


Optimality Condition May Hold for a Non-Optimal Consumption Bundle

Non-convex preferences



Non-convex budget set



Optimization in Case of n Goods

$$\max_{x_1, \dots, x_n} u(x_1, \dots, x_n) \quad \text{s.t.} \quad \text{(i)} \quad \sum_{i=1}^n p_i x_i \leq m, \quad \text{(ii)} \quad x_i \geq 0$$

If $u(x)$ is

- ① strictly quasiconcave (strictly convex preferences)
- ② continuous (continuous preferences)
- ③ nondecreasing (monotone preferences)

and set B is

- ① nonempty
- ② closed, bounded (compact)
- ③ convex

Then there is a unique solution

$$x^* : \sum_{i=1}^n p_i x_i^* = m$$

Optimization: Interior Solution

Lagrangian function:

$$\mathcal{L}(x_1, x_2, \dots, x_n, \lambda) = u(x_1, x_2, \dots, x_n) + \lambda(m - \sum_{i=1}^n p_i x_i)$$

First-order conditions (FOCs):

$$\frac{\partial \mathcal{L}(\mathbf{x}^*, \lambda^*)}{\partial x_i} = \frac{\partial u(\mathbf{x}^*)}{\partial x_i} - \lambda^* p_i = 0, \quad \forall i = 1, \dots, n$$

$$\frac{\partial \mathcal{L}(\mathbf{x}^*, \lambda^*)}{\partial \lambda} = m - \sum_{i=1}^n p_i x_i^* = 0$$

Rewriting FOCs:

$$\frac{\partial u(\mathbf{x}^*)}{\partial x_i} = \lambda^* p_i$$

Therefore, for any $i, j \in 1, \dots, n, i \neq j$:

$$\frac{\frac{\partial u(\mathbf{x}^*)}{\partial x_i}}{\frac{\partial u(\mathbf{x}^*)}{\partial x_j}} = \frac{\lambda^* p_i}{\lambda^* p_j} = \frac{p_i}{p_j} \quad \equiv \quad \text{Optimality condition}$$

Optimization: Lagrange Multiplier

What does λ^* express?

$$\lambda^* = \frac{\frac{\partial u(\mathbf{x}^*)}{\partial x_i}}{p_i} = \frac{MU_i}{p_i} = \frac{MU_j}{p_j} \quad \forall i, j \in \{1, 2, \dots, n\}$$

$$\frac{\partial \mathcal{L}(\mathbf{x}^*, \lambda^*)}{\partial m} = \sum_{i=1}^n \left(\frac{\partial u(\mathbf{x}^*)}{\partial x_i} \frac{\partial x_i}{\partial m} - \lambda^* p_i \frac{\partial x_i}{\partial m} \right) + \lambda^* = \lambda^*$$

λ expresses:

- utility value that can be obtained with the next dollar of consumption
= shadow price of the budget constraint
- technically – measures the effect of a marginal shift of the constraint border upon the optimum value of the objective function

Optimization: Example

Example:

$$u(x, y) = \frac{1}{4} \ln x + \frac{3}{4} \ln y$$

Is $u(x, y)$ continuous, monotonous and strictly quasiconcave?

Let $p_x = 1$, $p_y = 2$, $m = 12$: find optimal consumption bundle

Convenient property of Cobb-Douglas utility function $u(x, y) = x^a \cdot y^b$:

$$\frac{p_x x^*}{m} = \frac{a}{a+b}, \quad \frac{p_y y^*}{m} = \frac{b}{a+b}$$

Optimization: Corner Solution

$$x_j^* = 0 : \quad MRS_{ji}(\mathbf{x}^*) = -\frac{MU_i(\mathbf{x}^*)}{MU_j(\mathbf{x}^*)} \leq -\frac{p_i}{p_j}$$

$$\Rightarrow \frac{MU_i(\mathbf{x}^*)}{p_i} \geq \frac{MU_j(\mathbf{x}^*)}{p_j}$$

$$\mathcal{L}(x_1, x_2, \dots, x_n, \lambda) = u(x_1, x_2, \dots, x_n) + \lambda(m - \sum_{i=1}^n p_i x_i)$$

Kuhn-Tucker (complementary slackness) conditions:

For all $i = 1, \dots, n$,

1. $\frac{\partial \mathcal{L}(\mathbf{x}^*, \lambda^*)}{\partial x_i} = \frac{\partial u(\mathbf{x}^*)}{\partial x_i} - \lambda^* p_i \leq 0 \quad \wedge \quad x_i^* \geq 0 \quad \wedge \quad x_i^* \cdot \left(\frac{\partial u(\mathbf{x}^*)}{\partial x_i} - \lambda^* p_i \right) = 0;$
2. $\frac{\partial \mathcal{L}(\mathbf{x}^*, \lambda^*)}{\partial \lambda} = m - \sum_{i=1}^n p_i x_i^* = 0.$

The Carte Blanche Principle

The Carte Blanche principle

Consumers are always weakly better off receiving a cash transfer than an in-kind transfer of equivalent monetary value.

In-kind transfers: non-cash transfers of goods and services (e.g. food stamps, vouchers, subsidized educational loans, child care services, job training)

If consumers are rational, in-kind transfers (constraints on choice) cannot be beneficial when compared to cash transfers.

The Carte Blanche Principle

Gift giving – in-kind transfer — less efficient than giving cash

J. Waldfogel (1993). The Deadweight Loss of Christmas. *American Economic Review*, 83(5), 1328–1336.

Deadweight loss depends on the giver's knowledge of recipient's preferences

Survey of 150 Yale students about Christmas gifts

$$\text{Gift yield} = \frac{\text{Student's willingness to pay for the gift}}{\text{Price of the gift}}$$

Summary of findings:

- \$100 gift – recipient valuation estimated to \$62
- Value destruction greater for distant relatives (grandparents)
- Value preservation near-perfect for friends
- Groups that tend to destroy the most value are the most likely to give cash